



Practice: Textbook Exercise 1.6.7 (set size via bijections)

In everyday mathematics, the number of elements of a set is understood in an intuitive sense and no definition is usually given. In the logical foundations of mathematics, however, the number of elements is defined via bijections: $|X| = n$ means that there exists a bijection from X to the set $\{1, 2, \dots, n\}$. (Also other, alternative definitions of set size exist but we will consider only this one here.)

- Prove that if X and Y have the same size according to this definition, then there exists a bijection from X to Y .
- Prove that if X has size n according to this definition, and there exists a bijection from X to Y , then Y has size n too.
- * Prove that a set cannot have two different sizes m and n , $m \neq n$, according to this definition. Be careful not to use the intuitive notion of “size” but only the definition via bijections. Proceed by induction.

Answer: (a) Suppose $|X| = n$ and $|Y| = n$. By definition there exist bijections $f: X \rightarrow \{1, \dots, n\}$ and $g: Y \rightarrow \{1, \dots, n\}$. Then $g^{-1} \circ f: X \rightarrow Y$ is a composition of two bijections, hence itself a bijection from X to Y . $\square$

(b) Suppose $|X| = n$, so there exists a bijection $f: X \rightarrow \{1, \dots, n\}$. Let $g: X \rightarrow Y$ be a bijection. Then $f \circ g^{-1}: Y \rightarrow \{1, \dots, n\}$ is a composition of two bijections, hence a bijection, so $|Y| = n$ by definition. $\square$

(c) Suppose $|X| = n$ and $|X| = m$. By definition there exist bijections $f: X \rightarrow \{1, \dots, n\}$ and $g: X \rightarrow \{1, \dots, m\}$. Then $g \circ f^{-1}: \{1, \dots, n\} \rightarrow \{1, \dots, m\}$ is a bijection. It therefore suffices to prove that this forces $m = n$.

We show by *induction on n* that if there exists a bijection $h: \{1, \dots, n\} \rightarrow \{1, \dots, m\}$, then $m = n$.

Base case ($n = 1$). If $h: \{1\} \rightarrow \{1, \dots, m\}$ is a bijection then h is surjective, so every element of $\{1, \dots, m\}$ is hit. Since the domain has exactly one element, $m = 1$.

Inductive step. Assume the claim holds for $n - 1$: any bijection $\{1, \dots, n - 1\} \rightarrow \{1, \dots, m'\}$ forces $m' = n - 1$. Now suppose $h: \{1, \dots, n\} \rightarrow \{1, \dots, m\}$ is a bijection; we must show $m = n$.

We may assume WLOG that $h(n) = m$. Indeed, if $h(n) = k \neq m$, set $j = h^{-1}(m)$ and define $\tilde{h}$ by swapping the images of n and j :

$$\tilde{h}(i) = \begin{cases} h(i) & i \notin \{j, n\}, \\ m & i = n, \\ k & i = j. \end{cases}$$

Then $\tilde{h}$ is still a bijection $\{1, \dots, n\} \rightarrow \{1, \dots, m\}$ with $\tilde{h}(n) = m$.

Now restrict: the map $\tilde{h}|_{\{1, \dots, n-1\}}: \{1, \dots, n-1\} \rightarrow \{1, \dots, m-1\}$ is a bijection (removing the pair $n \mapsto m$ preserves bijectivity). By the inductive hypothesis $m - 1 = n - 1$, hence $m = n$. $\square$